Program 9 : Orgabnize Kubernetes pods using labels and namespaces

namespaces.yaml:

```
GNU nano 7.2                    namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod

^G Help       ^O Write Out   ^W Where Is   ^K Cut     ^T Execute   ^C Location
^X Exit       ^R Read File   ^\ Replace    ^U Paste   ^J Justify   ^/ Go To Line
```

pods.yaml:

```
GNU nano 7.2                    pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:

^G Help       ^O Write Out   ^W Where Is   ^K Cut     ^T Execute   ^C Location
^X Exit       ^R Read File   ^\ Replace    ^U Paste   ^J Justify   ^/ Go To Line
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
```

```
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

Deployment Execution

The configurations are applied to the Kubernetes cluster using the `kubectl apply` command.

- Create Namespaces:
  The command `kubectl apply -f namespaces.yaml` is used to create the required namespaces, namely dev and prod, in the Kubernetes cluster.
- Create Pods:
  The command `kubectl apply -f pods.yaml` deploys the specified pods such as frontend-pod and backend-pod into their respective namespaces (dev and prod) as defined in the YAML configuration file.

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods
No resources found in default namespace.
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods
No resources found in default namespace.
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods -n dev
NAME           READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          52s
frontend-pod   1/1     Running   0          52s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods -n prod
NAME           READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          58s
frontend-pod   1/1     Running   0          58s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods -A
NAMESPACE     NAME                               READY   STATUS    RESTARTS        AGE
dev           backend-pod                        1/1     Running   0               67s
dev           frontend-pod                       1/1     Running   0               67s
kube-system   coredns-66bc5c9577-rvm7r           1/1     Running   2 (4m12s ago)   102m
kube-system   etcd-minikube                      1/1     Running   2 (4m16s ago)   102m
kube-system   kube-apiserver-minikube            1/1     Running   2 (4m16s ago)   102m
kube-system   kube-controller-manager-minikube   1/1     Running   2 (4m17s ago)   102m
kube-system   kube-proxy-6x8mx                   1/1     Running   2 (4m17s ago)   102m
kube-system   kube-scheduler-minikube            1/1     Running   2 (4m17s ago)   102m
kube-system   storage-provisioner                1/1     Running   5 (69s ago)     102m
prod          backend-pod                        1/1     Running   0               67s
prod          frontend-pod                       1/1     Running   0               67s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$
```

Verification of Pods

After deploying the namespaces and pods, the pod status was verified using various `kubectl get pods` commands.

- Check pods in the default namespace:
  The command `kubectl get pods` showed running pods related to the nginx deployment in the default namespace.
- Check pods in the dev namespace:
  The command `kubectl get pods -n dev` displayed the frontend-pod and backend-pod, both in the Running state, confirming successful deployment in the dev namespace.
- Check pods in the prod namespace:
  The command `kubectl get pods -n prod` confirmed that the frontend-pod and backend-pod were also running successfully in the prod namespace.
- Filter pods using labels:
  The command `kubectl get pods -n dev -l app=frontend` listed only the frontend pod.
  The command `kubectl get pods -n dev -l app=backend` listed only the backend pod.
  This verifies that labels were correctly assigned and used to organize pods.

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0          3m41s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          3m56s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/labb9$ 
```